# Ex - 08
# Data Visualization Using Tableau

## AIM

To visualize data using Tableau by connecting to various data sources, creating bar charts, line charts, and pie charts, implementing calculated fields, and building interactive dashboards and stories using the T20 Cricket dataset.

## PROBLEM STATEMENT

To analyze and visualize the T20 Cricket dataset using Tableau by connecting Excel, CSV, and SQL databases, creating basic charts, performing calculations, and developing interactive dashboards to present meaningful insights such as top run scorers and match winners.

## PROCEDURE

1. Open Tableau Desktop and explore the interface, including the Data Source page, worksheet, Rows shelf, Columns shelf, Marks card, and Show Me panel.
2. Connect to the T20 Cricket dataset (Excel or CSV) by selecting the appropriate option under Connect, choosing the file, and loading the required data.
3. Connect to a SQL database by selecting the required database connector, entering the server details, providing authentication if needed, and selecting the required tables.
4. Check the imported data, verify the data types, rename fields if necessary, and remove unwanted columns or records to prepare the dataset for visualization.
5. Create a bar chart by dragging Batsman Name to Rows and Runs to Columns, then select Bar Chart from the Show Me panel and sort to show the top 10 run scorers.
6. Create a line chart by dragging a date field to Columns and a numerical field (such as Runs) to Rows, then select Line from the Marks card to display trends over time.
7. Create a pie chart by selecting Pie from the Marks card, dragging a category field (such as Winner) to Color, and dragging a numerical field to Angle to display proportions.
8. Create a calculated field by selecting Analysis > Create Calculated Field, entering a formula, such as COUNT([Match Id]), and clicking OK to generate a new calculated field.
9. Build a dashboard by selecting New Dashboard, dragging the required worksheets into the dashboard, and arranging charts, titles, legends, and filters to create an interactive layout.
10. Create a story by selecting New Story, adding worksheets or dashboards as story points, providing captions, and saving the workbook for further analysis and presentation.

## IMPLEMENTATION

1. Calculated Field - Matches Won

```
COUNT([Match Id])
```

2. Calculated Field - Run Category

```
IF [Runs] >= 200 THEN "High Scorer"
ELSE "Low Scorer"
END
```

Dashboard components:

• Bar chart - Top 10 run scorers (Batsman Name vs Runs)

• Gantt/bar chart - Match-wise winners (Winner vs Match Id)

• Filter - Select a team, batsman or match

• Story - Present visualizations in a sequence

## OUTPUT

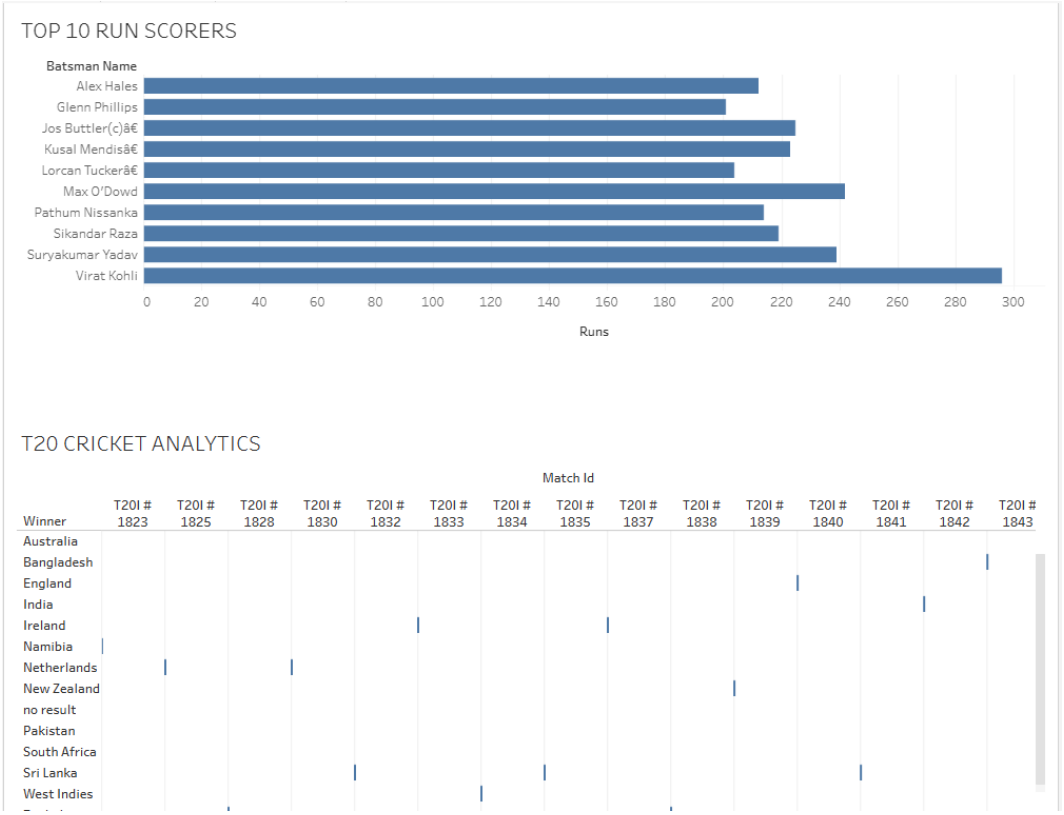


*Fig: T20 Cricket Analytics Dashboard*

## RESULT

Data visualization using Tableau was successfully studied and implemented on the T20 Cricket dataset. Data was connected from different sources, bar charts, line charts, and pie charts were created, calculated fields were implemented, and interactive dashboards and stories were developed to analyze and present meaningful insights.